JavaScript:

Sr. No.	Area	Topic	What to Cover	Notes / Context
1	Foundations	What is JavaScript	Role of JS, where it runs, use cases	Browser and Server
2	Foundations	JavaScript Engines	How JS is executed	V8, SpiderMonkey
3	Syntax	Basic Syntax	Statements, expressions, comments	Language basics
4	Syntax	Variables	var vs let vs const	Scope and mutability
5	Data Types	Primitive Types	string, number, boolean, null, undefined, symbol, bigint	Type system basics
6	Data Types	Reference Types	Object, Array, Function	Heap vs stack concept
7	Operators	Operators	Arithmetic, comparison, logical, ternary	Type coercion awareness
8	Control Flow	Conditional Statements	if, else, switch	Branching logic
9	Control Flow	Loops	for, while, do-while, for...of	Iteration patterns
10	Functions	Function Declaration	Named and anonymous functions	Reusability
11	Functions	Arrow Functions	Syntax differences, this binding	ES6 features
12	Scope	Scope Rules	Global, function, block scope	let/const behavior
13	Scope	Hoisting	Variable and function hoisting	Common pitfalls
14	Objects	Object Basics	Properties, methods, access patterns	Key–value model
15	Objects	Object Mutation	Reference behavior	Side effects
16	Arrays	Array Basics	Creation, indexing, iteration	Core operations
17	Arrays	Array Methods	map, filter, reduce, forEach	Functional patterns
18	Equality	Equality Checks	== vs ===	Type coercion risks

19	Error Handling	Errors and Exceptions	try, catch, finally	Runtime safety
20	Runtime	Call Stack	How JS executes code	Single-threaded model
21	Runtime	Event Loop	Asynchronous execution model	Conceptual understanding
22	Async	Callbacks	Callback pattern and issues	Callback hell
23	Async	Promises	then, catch, chaining	Async control
24	Async	async / await	Synchronous-looking async code	Error handling
25	Timing	Timers	setTimeout, setInterval	Event loop interaction
26	This Keyword	this Binding	Context rules	Function vs arrow
27	Prototypes	Prototype Chain	Inheritance model	Object behavior
28	Classes	ES6 Classes	class syntax, constructors	Syntactic sugar
29	Modules	JS Modules	import, export	Code organization
30	Standard Library	Math, Date, JSON	Built-in utilities	Practical usage
31	Browser Basics	DOM Concept	What the DOM is	Browser-only
32	Browser Basics	DOM Manipulation	Querying and updating elements	High-level
33	Browser Basics	Events	Event handling basics	addEventListener
34	Debugging	Debugging JS	Console, breakpoints	Developer tools
35	Memory	Memory Management	Garbage collection concept	Leak awareness
36	Best Practices	Immutability	Avoiding unintended mutation	Predictability
37	Best Practices	Defensive Coding	Null/undefined checks	Runtime safety
38	Security	Common JS Risks	XSS basics, unsafe eval	Awareness only
39	Environment	Browser vs Node.js	Global objects, APIs	Conceptual difference
40	Coding Discipline	Clean JavaScript	Readability, naming, structure	Maintainability

41	Web Communication	What is AJAX	Meaning of AJAX, problem it solves	Asynchronous requests
42	Web Communication	HTTP Basics for AJAX	Methods, status codes, headers	REST fundamentals
44	Web Communication	Fetch API	Modern AJAX approach	Promise-based
45	Web Communication	Fetch vs XHR	Why Fetch replaced XHR	Simpler API
46	Web Communication	Sending Data	Query params vs request body	GET vs POST
47	Web Communication	JSON Handling	Serializing and parsing JSON	JSON.stringify / parse
48	Web Communication	Handling Responses	Success, error, and edge cases	Status vs network errors
49	Web Communication	Error Handling in AJAX	HTTP errors, timeouts	try/catch with async
51	Web Communication	Cookies & Credentials	Sending cookies with requests	credentials: include
53	Web Communication	Loading States	UI feedback during requests	User experience
54	Web Communication	Retry Logic	Handling transient failures	Idempotent requests
55	Web Communication	Parallel Requests	Multiple AJAX calls	Promise.all
56	Web Communication	Sequential Requests	Dependent API calls	await chaining
57	Web Communication	Canceling Requests	Aborting ongoing requests	AbortController
58	Web Communication	File Upload via AJAX	Multipart form data	FormData API
59	Web Communication	File Download via AJAX	Handling blobs	Response.blob()

60	Best Practices	API Contract Discipline	Frontend–backend alignment	DTO consistency